

Principles of Drug Therapy, Dosing, and Prescribing in Chronic Kidney Disease and Kidney Replacement Therapy

Matthew J. Cervelli, Terry E. Jones, Sanja Mirkov

Reduced glomerular filtration rate (GFR) can alter drug pharmacokinetics and, consequently, patients with chronic kidney disease (CKD) are at greater risk of adverse effects. In addition, these patients take multiple drugs and are at higher risk for drug interactions and drug-related problems.¹ To prescribe safely and effectively, clinicians should be familiar with the pharmacokinetics of drugs in varying stages of CKD and kidney replacement therapy (KRT) and ideally rely on data from these populations. Unfortunately, such information is not always available, and exclusion of these patients from clinical studies can lead to restrictive recommendations. This chapter describes pharmacokinetic principles and highlights common prescribing issues in patients with CKD, dialysis, and transplantation. Specific dose recommendations and pharmacokinetic data are not included but can be obtained elsewhere.^{2–5}

PHARMACOKINETIC PRINCIPLES

Pharmacokinetics describes the behavior of a drug (or metabolite) in the body in terms of its absorption, distribution, metabolism, and elimination (Fig. 77.1 and Table 77.1).^{6,7}

Absorption

Bioavailability

Bioavailability (F) is the portion of the drug dose that appears in the systemic circulation after administration. Drugs given intravenously have 100% bioavailability, whereas drugs given by alternative routes pass through a series of biologic membranes before entering the systemic circulation so that only a fraction may reach the circulation. After oral administration, the liver can metabolize a drug during first pass as it is being absorbed or later as it is delivered via systemic blood flow. First-pass metabolism can significantly reduce absorption. The gastrointestinal (GI) mucosa can also influence absorption by the presence of drug-metabolizing enzymes and drug efflux pumps including P-glycoprotein.⁸

CKD can theoretically influence drug absorption, although these effects are difficult to quantify, and clinical examples are limited. GI edema can limit oral drug absorption (e.g., furosemide in nephrotic syndrome). Nausea and vomiting from uremia can impair absorption and contact time between a drug and GI mucosa. In advanced uremia, the alkalinizing effect of salivary urea may decrease absorption of drugs

optimally absorbed in an acid milieu. Commonly prescribed metallic divalent ion phosphate binders (e.g., aluminum, calcium, and magnesium salts) can decrease drug absorption by forming nonabsorbable complexes with drugs (Table 77.2).⁹ Changes in cardiac output can reduce the rate and extent of absorption for drugs with significant first-pass metabolism.

Increased absorption in patients with CKD from reduced first-pass metabolism is seen with some β -blockers, dextropropoxyphene, and dihydrocodeine. Comorbidities in CKD patients also can have an effect (e.g., erratic absorption because of diabetic GI neuropathy).

Distribution

Volume of Distribution

After absorption, drugs may distribute from plasma to an extravascular compartment. Each drug has a characteristic volume of distribution (V_D), which does not correspond to an anatomic space but instead relates the amount of drug in the plasma to the amount in tissue. A high V_D means a relatively smaller amount remains in plasma and more is in tissues. V_D is used to calculate the loading dose to achieve a desired plasma concentration ($V_D = \text{dose}/[\text{plasma}]$). Water-soluble drugs tend to be restricted to the extracellular fluid space and have a relatively small V_D . Lipid-soluble drugs penetrate body tissues and have a large V_D . Increases in V_D can occur with edema, ascites, or infection, particularly for water-soluble drugs. If usual doses are given, low concentrations result. Conversely, muscle wasting or volume depletion can decrease the V_D of water-soluble drugs, and usual doses produce high concentrations.

Plasma Protein Binding

Some drugs bind extensively to plasma proteins.¹⁰ It is only the free (unbound) fraction that exerts a pharmacologic effect. If protein binding is reduced, a greater free fraction is available for any given total drug concentration, which may increase drug activity. Organic acids usually have a single binding site on albumin, whereas organic bases have multiple binding sites on glycoproteins. Protein binding can be altered in CKD, especially when serum albumin is low (e.g., nephrotic syndrome) or when uremic toxins displace drugs from binding sites (Table 77.3). Predicting the effect of changes in protein binding is difficult and effects may be transient after acute displacement. For example, phenytoin has marked decreases in protein binding in patients

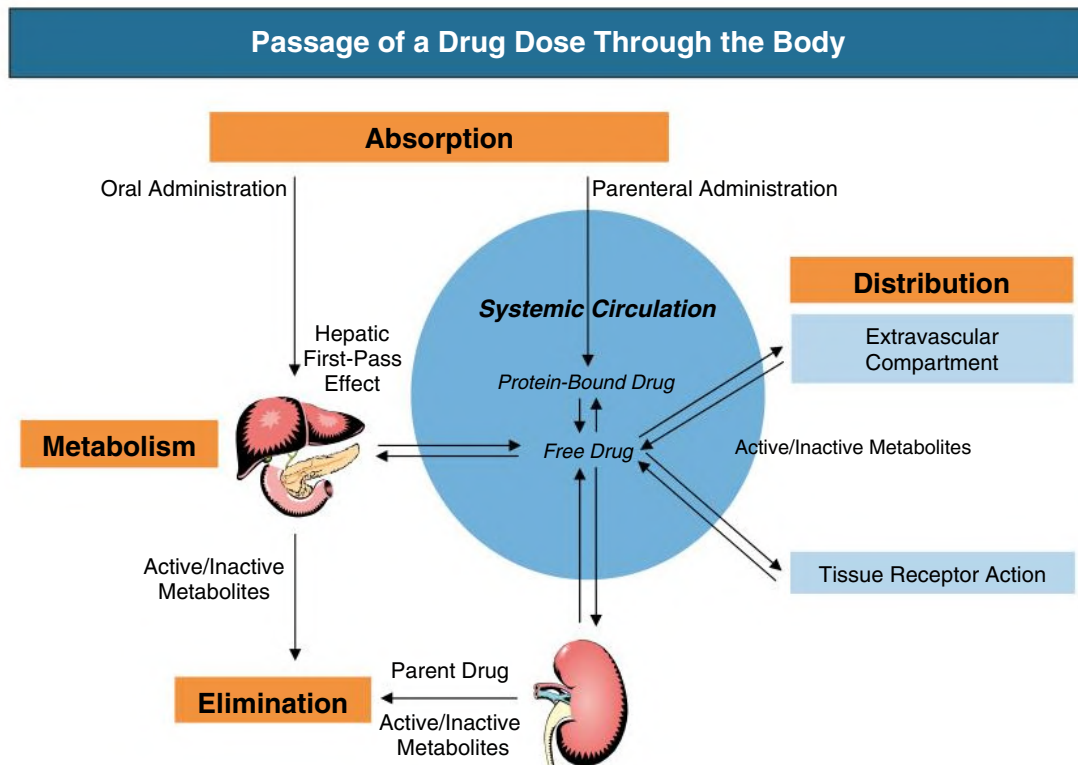


Fig. 77.1 Passage of a drug dose through the body.

TABLE 77.1 Pharmacokinetic Parameters

Parameter	Definition	Application
Bioavailability (<i>F</i>)	Percentage of a dose that appears in the systemic circulation after administration.	Determines the amount of drug reaching the systemic circulation. Determines intravenous and oral dose sizes.
Volume of distribution (V_D)	Proportionality constant relating the amount of drug in the body at a given time to a simultaneously occurring drug concentration in plasma or blood. A primary pharmacokinetic parameter.	Determines the size of loading doses. Explains the percentage of the total drug residing in plasma.
Clearance	Volume of blood cleared of the drug in a unit of time; a primary pharmacokinetic parameter.	Determines the “daily” maintenance dose and frequency of dosing.
Half-life ($t_{1/2}$)	Time taken for the drug concentration in plasma to fall to half its current value. A derived parameter calculated from both clearance and V_D .	Determines the frequency of administration and time to steady state.

with CKD, and toxicity can occur despite normal or low total plasma concentrations because of an increase in the free fraction. With albuminuria, bound drug may also be lost, which partially explains the refractoriness of nephrotic patients to diuretics. In CKD, high plasma levels of α_1 -acid glycoprotein are induced in acute and chronic inflammation, which can increase drug binding.

Metabolism

Drug metabolism is primarily a hepatic function by which drugs are converted to more water-soluble entities, which promotes elimination by the kidneys and bile. Despite the assumption that extra renal clearance is unchanged, reduced GFR can alter and slow drug metabolism.¹¹ Importantly, some drugs have active or toxic metabolites that are cleared by the kidney and such metabolites can accumulate in patients with CKD. The kidneys also have a metabolic role with insulin and imipenem.

Elimination

The kidney is the most important organ for drug and metabolite elimination. Terms to describe drug clearance are shown in Box 77.1. Total drug clearance is expressed as the volume of blood or plasma from which the drug is cleared per unit of time and can be calculated by dividing dose by the area under the drug concentration curve (AUC). *Half-life* describes the time taken for plasma concentrations to halve and is affected by both V_D and clearance. Quantitation of drug elimination by the kidney is expressed as kidney clearance, which depends on renal blood flow and the innate ability of the kidney to remove the drug. Renal drug clearance is the sum of its removal by GFR and tubular secretion, minus that reabsorbed via tubular reabsorption. Glomerular filtration depends on molecular size (<10 kDa), charge, and free fraction. Tubular secretion of drugs may change with kidney disease, but measurement of tubular function is difficult. As GFR decreases, drugs dependent on tubular secretion are also excreted more

TABLE 77.2 Effect of Food and Phosphate Binders on Oral Drug Absorption

Drug	Effect of Food
Captopril	Decreases serum drug levels
Bisphosphonates (oral)	Significantly reduce drug absorption
Cinacalcet	Significantly increases drug absorption
Iron (oral)	Decreases absorption
Ketoconazole or itraconazole	Increase absorption with reduced pH
Sirolimus	High-fat meals increase absorption
Tacrolimus	Reduces drug absorption
Bisphosphonates (oral)	Calcium-containing binders significantly reduce absorption
Fluoroquinolones	Reduction in absorption
Tetracycline	Reduction in absorption
Thyroid hormones	Reduction in absorption

BOX 77.1 Mathematics of Drug Elimination

Total body clearance = Drug dose/AUC

Renal clearance = Total amount of drug in urine/plasma drug concentration

Total amount of drug in urine = [Drug] × Volume of the sample collected in a fixed time

Drug half-life ($t_{1/2}$) = $V_D \times 0.693/\text{Clearance}$

AUC, Area under the concentration-time curve; V_D , volume of distribution (dose/blood concentration).

slowly, although tubular secretion may be relatively spared and total kidney clearance may be much higher than would be predicted from estimating GFR alone (e.g., penicillins). For a given level of extra renal clearance, as GFR falls, clearance of drugs (and metabolites) eliminated by the kidney also falls and half-life is prolonged.

PRESCRIBING PRINCIPLES FOR CHRONIC KIDNEY DISEASE AND KIDNEY REPLACEMENT THERAPY

Ideally, drug doses should be modified based on published data on drug kinetics in varying stages of CKD. However, lack of reliable data and individual patient factors limit this.¹² Dose nomograms, tables, and computer recommendations can guide dose modifications, but they are only a guide and therapeutic drug monitoring and clinical acumen are important too.

Initial Assessment and Laboratory Data

A targeted history is important in assessing dose in CKD patients. Previous drug efficacy or toxicity should be determined and the current prescribed medications reviewed for potential interactions or nephrotoxins. Physical and laboratory parameters indicate volume status, height, weight, and extrarenal disease (e.g., liver).

Estimating Renal Function for Drug Dosage

Estimating kidney function is essential to guide drug dosing in patients with CKD. The Cockcroft-Gault equation estimates creatinine clearance and is individualized for each patient (by body weight) and has been the most widely used and accepted method for drug dosage calculation. The Modification of Diet in Renal Disease (MDRD) formula and Chronic Kidney Disease–Epidemiology Collaboration (CKD-EPI)

TABLE 77.3 Protein Binding of Drugs in Renal Disease

Major Effects	Minor Effects
Albumin: Binding Sites for Acidic Compounds	
Barbiturates (↓)	Ascorbic acid (↑)
Benzodiazepine (↑)	Valproate (↓)
Carbamazepine (↑)	Fatty acids (↑)
Fibrates (↑)	Nafcillin (↑)
Furosemide (↓)	Phenylbutazone (↓)
Mycophenolate mofetil (↑)	Probenecid (↑)
Penicillins (↑)	Thiopental (↓)
Phenytoin (↓)	Warfarin (↓)
Sulfonamides (↓)	Thyroxine (↓)
Globulins: Binding Site for Basic Compounds	
Digoxin (↓)	Adenosine (↑)
Methadone (↑)	Amitriptyline (↑)
Propranolol (↑)	Chloramphenicol (↑)

↓ Indicates reduced protein binding in chronic kidney disease (CKD) patients; ↑ indicates increased protein binding in CKD patients. However, the therapeutic effect is not easily predicted (see text).

formula can give different estimates of kidney function, especially if not corrected for body surface area because they are not individualized and are normalized for body surface area ($\text{mL/min}/1.73 \text{ m}^2$). These equations can lead to recommendations different from those obtained by the Cockcroft-Gault equation. An important limitation of all kidney function estimates is inaccuracy of single-point estimates when kidney function is rapidly changing (and creatinine concentration is not at steady state). This may lead to overestimation or underestimation of kidney function and underdosing or overdosing. In patients with rapid-onset severe acute kidney injury (AKI), the decline in GFR is so rapid that patients may require doses for GFR below 10 mL/min. The opposite is true in rapidly improving kidney function after AKI or early posttransplant where there is a risk of underdosing.

Whether estimated using creatinine clearance or GFR, the greater the degree of reduction in kidney function, the greater the potential need for dose modification. With exceptions, dose modification usually is not clinically necessary until GFR (or creatinine clearance) is less than 30 mL/min/ 1.73 m^2 . Assessment of kidney function for drug dosing is not a precise science. What is important for clinical decision making is awareness that kidney function is impaired, and approximately to what extent, rather than knowledge of the precise GFR.¹³ Using the extent of reduction in GFR to classify CKD severity is discussed in [Chapters 3 and 80](#).

Activity and Toxicity of Metabolites

It is important to consider the activity and/or toxicity of drug metabolites in addition to that of the parent drug when determining drug selection and dosing in patients with CKD because metabolites that are normally cleared by the kidney can accumulate, leading to enhanced drug action or toxicity ([Table 77.4](#)).

Fraction of Active Drug (and Active or Toxic Metabolite) Excreted Unchanged in Urine

The greater the fraction of active drug or metabolite excreted unchanged by the kidneys (f_e), the greater is the need for dose modification. It is usually only clinically necessary to modify doses if the f_e is greater than 25% to 50%. When ascertaining f_e , it is important to note that values are often obtained from studies that do not distinguish between parent

TABLE 77.4 Drugs With Active or Toxic Metabolites That Accumulate in Patients With Reduced Glomerular Filtration Rate

	Active Metabolite	Consequence
Allopurinol	Oxypurinol	Nil
Cefotaxime	Desacetyl cefotaxime	
Glyburide	4- <i>trans</i> -Hydroxyglibenclamide 3- <i>cis</i> -Hydroxyglibenclamide	Hypoglycemia
Morphine	Morphine-6-glucuronide	CNS side effects
Tramadol	O-Desmethyltramadol	CNS side effects
Venlafaxine	O-Desmethylvenlafaxine	CNS and cardiovascular side effects
	Toxic Metabolite	Consequence
Dapsone	Monoacetylated metabolite	
Meperidine (pethidine)	Normeperidine (norpethidine)	CNS (seizures)
Nitroprusside	Thiocyanate	Cyanide toxicity
Procainamide	N-Acetylprocainamide (NAPA)	Arrhythmia
Propoxyphene	Norpropoxyphene	Cardiac toxicity

CNS, Central nervous system.

Fractional Drug or Metabolite Excretion in Urine

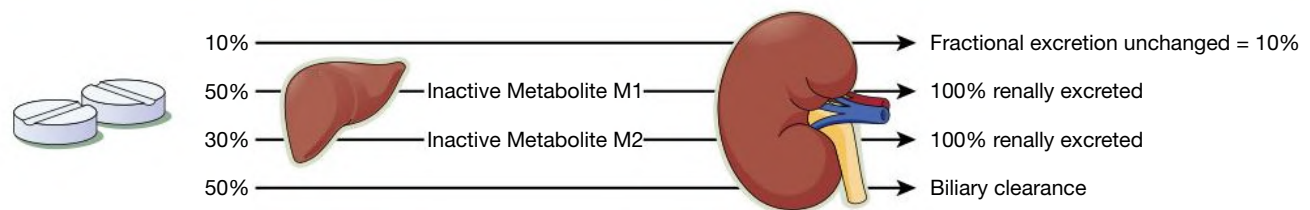


Fig. 77.2 Fraction of Drug or Metabolite Excreted in Urine. In this hypothetical example, 10% of the dose is excreted unchanged in urine ($f_e = 10\%$); 50% of the dose is metabolized to inactive metabolite M1, which is then all renally excreted; 30% of the dose is metabolized to inactive metabolite M2, which is all renally excreted; and the remaining 10% is excreted unchanged in bile. For this drug, the total excretion of drug dose in urine is 90%. However, this 90% includes 10% as parent drug and 80% as inactive metabolites, and dose modification is probably not essential even in severe chronic kidney disease. Total kidney excretion of the dose is 90%; however, the clinically significant fraction of active drug excreted in urine is 10%.

drug and metabolites. The contribution of inactive/nontoxic metabolites to overall kidney drug elimination may exaggerate the potential for harm. Active or toxic metabolites should be assessed separately for their dependence on kidney elimination in the same way as the parent drug (Fig. 77.2).

Therapeutic Index of the Drug or Metabolites

The decision to modify dosage in CKD patients is influenced by the therapeutic window or index of the drug. The *therapeutic window* is the range of plasma drug concentrations between the minimum concentration for clinical efficacy and minimum concentration for toxicity. The *therapeutic index* is the ratio of these concentrations (Fig. 77.3). If the therapeutic window is wide (e.g., many penicillins), there may be no clinical need for dose modification despite much higher plasma concentrations found in CKD patients. If the therapeutic window is narrow (e.g., digoxin), dose modification is more important. Clinicians should judge the clinical relevance of increased exposure to drug or metabolites.

Avoiding Nephrotoxic Drugs

A wide range of drugs can cause nephrotoxicity (Table 77.5). Idiosyncratic nephrotoxicity (e.g., interstitial nephritis; see Chapter 64) is unpredictable and independent of dose. Predictable hemodynamic-related nephrotoxicity can occur with any drug that reduces blood

Therapeutic Index of a Drug

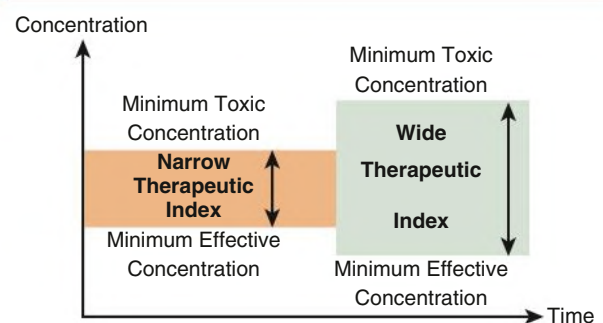


Fig. 77.3 Therapeutic Index of a Drug. Example of a drug with a narrow therapeutic index (left), and a drug with a wide therapeutic index (right).

volume, systemic blood pressure, or glomerular pressure because these affect the amount of blood delivered to the kidneys. These include all antihypertensives but especially angiotensin-converting enzyme (ACE) inhibitors and angiotensin receptor blockers (ARBs), nonsteroidal anti-inflammatory drugs (NSAIDs), diuretics, and laxatives. Many of these compounds result in a decrease in estimated GFR (eGFR) without structural injury and the eGFR reverses on stopping the drug

TABLE 77.5 Examples of Nephrotoxic Drugs

Examples	Mechanism	Prevention and Management
ACE inhibitors, ARBs	Impair angiotensin II–mediated afferent arteriole dilation during kidney hypoperfusion	Withdraw in kidney hypoperfusion (e.g., extracellular volume depletion)
Aminoglycosides (amikacin, gentamicin, tobramycin)	In proximal tubules, aminoglycosides bind to anionic phospholipid, are delivered to megalin, are taken up into the cell, accumulate, and cause direct toxicity.	Alternative if possible Monitor drug concentrations Avoid multiple daily doses Withdraw if SCr rises
Antifungals (amphotericin)	Afferent vasoconstriction and direct action to reduce GFR Distal tubular injury via creation of pores that increase membrane permeability leading to hypokalemia, hypomagnesemia, metabolic acidosis caused by tubular acidosis, polyuria from nephrogenic diabetes insipidus	Avoid use Administer slowly with hydration Use liposomal preparations
Antivirals Acyclovir	High drug levels result in drug crystals causing intratubular obstruction and foci of interstitial inflammation	Avoid bolus dose Reduce dose in CKD Hydrate during therapy
Cidofovir	Induces apoptosis in proximal tubule leading to tubular dysfunction, diabetes insipidus, kidney failure	Oral probenecid and hydration
Foscarnet	Direct tubular toxicity leading to acute tubular necrosis, nephrogenic diabetes insipidus Crystals in glomerular capillary lumen and proximal tubular lumen	Hydration
Indinavir	Crystal neuropathy, nephrolithiasis leading to obstructive AKI	Hydration
Calcineurin inhibitors (cyclosporine, tacrolimus)	↓ PG and ↑ 20-HETE acid production leading to vasoconstriction, generation of H ₂ O ₂ resulting in depleted glutathione leading to decreased GFR, ischemic collapse or scarring of the glomeruli, vacuolization of the tubules, and focal areas of tubular atrophy and interstitial fibrosis	Measure plasma concentrations Avoid interacting drugs Withdraw drug (switch to mTOR inhibitor)
Chemotherapeutics Cisplatin	<i>Cis</i> chloride replaced by H ₂ O leading to highly reactive OH radical leading to DNA injury, tubular cell death, nephrogenic diabetes insipidus, hypomagnesemia (may be persistent)	Forced diuresis and hydration
Ifosfamide	Direct tubular injury and mitochondrial damage leading to kidney tubular acidosis, Fanconi-like syndrome, nephrogenic diabetes insipidus, hypokalemia	
Intravenous immunoglobulin (sucrose-containing products)	Accumulation of sucrose in proximal convoluted tubules Increased osmolarity leading to cell swelling, vacuolization, and tubular luminal occlusion	Infusion rate < 3 mg sucrose/kg/min Avoid radioccontrast Avoid sucrose-containing product Hydration
Lithium	Impairment of collecting duct concentrating ability leading to diabetes insipidus Chronic tubulointerstitial nephropathy (tubular atrophy and interstitial fibrosis)	Measure plasma concentrations Prevent dehydration Avoid thiazides Avoid NSAIDs
NSAIDs	Hemodynamically induced AKI caused by vasoconstriction via reduced prostaglandin production Recruitment and activation of lymphocytes leading to acute and chronic tubulointerstitial nephritis, with or without nephrotic syndrome	Avoid use Withdraw during hypoperfusion Withdraw (add corticosteroids)
Proton pump inhibitors	Allergic acute interstitial nephritis	Withdraw (add corticosteroids)
Radioccontrast media	High osmolarity, medullary vasoconstriction, ↑ active transport in thick ascending loop of Henle leading to ↑ O ₂ demand	Hydration preprocedure and postprocedure acetylcysteine
Sulfonamides	Intrarenal precipitation leading to kidney stone formation	Fluid intake > 3 L/day; monitor urine for crystals Alkalize urine to pH > 7.15 if crystals seen

ACE, Angiotensin-converting enzyme; AKI, acute kidney injury; ARBs, angiotensin receptor blockers; CKD, chronic kidney disease; GFR, glomerular filtration rate; 20-HETE acid, 20-hydroxyeicosatetraenoic acid; H₂O₂, hydrogen peroxide; mTOR, mammalian target of rapamycin; NSAIDs, nonsteroidal anti-inflammatory drugs; PG, prostaglandin; SCr, serum creatinine.

(especially sodium-glucose cotransporter-2 [SGLT2] inhibitors and ACE inhibitors), but if perfusion pressure is sufficiently reduced, ischemic injury might occur. Direct tubular nephrotoxins include aminoglycosides, vancomycin, amphotericin, cisplatin, calcineurin inhibitors (CNIs), and radiographic contrast media. Obstructive uropathy can occur with crystallization of acyclovir or creatinine phosphokinase from statin-induced rhabdomyolysis. In dialysis patients with no significant residual kidney function, use of nephrotoxic drugs may be acceptable. Drug nephrotoxicity is discussed further in [Chapter 70](#).

Drugs That Aggravate the Metabolic/Biochemical Effects of Reduced Glomerular Filtration Rate

Some drugs have no direct adverse effect on kidney function but when used in CKD patients can aggravate the metabolic/biochemical consequences of reduced GFR. Hyperkalemia is exacerbated by potassium supplements, potassium-sparing diuretics, aldosterone antagonists, and blockers of the renin-angiotensin system. The catabolic effects of tetracycline can exacerbate uremia. Sodium-containing drugs and those that promote sodium and water retention (e.g., corticosteroids) should be used cautiously because they may provoke fluid overload and hypertension.

Effect of Reduced Glomerular Filtration Rate on Pharmacodynamic or Physiologic Mechanisms

CKD may alter a pharmacodynamic response or physiologic process, which, in turn, affects clinical response. For example, the inability of impaired kidneys to activate vitamin D precursors means that vitamins D₂ and D₃ may be less effective. Also, CKD patients often have a coagulopathy from the effects of uremia on platelet function and may be more prone to the bleeding complications of anticoagulant and antiplatelet therapy.

Effect of Reduced Glomerular Filtration Rate on the Concentration of Drug at the Site of Action

Reduced GFR can alter drug concentration at the site of action. Some diuretics (furosemide) and antibiotics (nitrofurantoin) are less effective in CKD patients because they do not achieve adequate concentrations at their site of action within the lumen of the kidney tubules (furosemide) or bladder (nitrofurantoin).

Location of Drug Action

Drugs that have negligible bioavailability and those that are used for a local or topical effect may be given safely at normal dose despite toxicity with systemic doses. These include topical NSAIDs, nebulized gentamicin, and oral vancomycin.

Method of Administration

In fluid-restricted patients, administration of intravenous (IV) drug infusions with approved fluid volumes may be undesirable. When the volume needed to administer IV drugs impacts on daily fluid restrictions, consider alternative drugs or more concentrated solutions, provided that physiochemical parameters allow. Similarly, oral drug administration that requires large fluid volumes (e.g., bisphosphonates) may be problematic. In patients with severe nausea and vomiting, “essential” drugs (e.g., immunosuppressants) should be administered intravenously.

Drug Interactions

Pharmacokinetic drug interactions are frequently problematic, and awareness of clinically significant interactions is essential, especially in patients taking transplant immunosuppressants. The most important of these are cyclosporine, tacrolimus, everolimus, and

sirolimus, which are substrates of both the cytochrome P450 (3A4) drug metabolizing enzyme system and P-glycoprotein drug efflux system that is expressed in GI mucosa, kidney, and liver ([Fig. 77.4](#)).¹⁴ Coprescription of drugs that inhibit these systems (e.g., some azole antifungals, calcium channel blockers, macrolides, and grapefruit juice) can increase absorption and reduce clearance of the immunosuppressant, causing toxicity. Conversely, drugs that induce these systems (e.g., barbiturates, phenytoin, carbamazepine, rifampicin, and St. John’s wort [*Hypericum*]) can reduce absorption and increase clearance and therefore increase the risk for rejection (see [Fig. 77.4](#)). All drug changes in patients receiving transplant immunosuppression should be considered for their potential to interact, and appropriate dose modifications or alternatives should be used. In addition, enzymes in the epithelial cells of the proximal tubules show an activity similar to the corresponding liver enzymes. Therefore, the kidneys can produce inactive, active, or nephrotoxic metabolites (e.g., metabolites of paracetamol).

Clinical Condition of the Patient

Patient welfare should override theoretical concerns. Higher-than-recommended doses may be appropriate when there is a strong clinical indication. For example, excessive reduction in initial antibiotic doses based on kidney function may be inappropriate in patients with life-threatening infection when the consequences of failed therapy are greater than those of potential toxicity.

METHODS OF DOSE REDUCTION IN CHRONIC KIDNEY DISEASE PATIENTS

Loading Doses

In general, steady-state drug concentrations are achieved after five drug half-lives. Steady-state can be achieved “earlier” by administering loading doses. If an appropriate (i.e., adjusted for GFR) dose is administered from the outset, attainment of steady state will be further prolonged (because half-life is longer). When a drug equilibrates rapidly with the tissues it diffuses into, the loading dose (mg/kg) is equal to the product of the desired plasma concentration (mg/mL) and V_D (mL/kg) and is independent of clearance. If a drug equilibrates slowly with the tissues it diffuses into, several smaller loading doses will be needed (e.g., vancomycin loading doses rarely achieve steady state concentrations after the first loading dose). Provided the desired concentration and V_D are unchanged, loading doses do not require modification in CKD patients. If V_D increases and it is important to achieve a desired plasma concentration quickly, then a larger loading dose might be needed even though the affected drug is renally cleared (and hence subsequent, maintenance doses are lower than normal).

Maintenance Doses

When specific pharmacokinetic information is not available, and assuming no change in nonrenal clearance, maintenance doses should be reduced in proportion to the extent of reduction in GFR. For example, if kidney function is 50% of normal and the drug is 100% renally excreted, a maintenance dose of 50% is required. If the drug is 50% cleared by the kidney and GFR is 20% of normal, the dose should be 60% of normal. The dose reduction factor is estimated from first principles or the following formula:

$$\text{Fraction of normal kidney function dose to use in kidney impairment} = 1 - f_e (1 - \text{Fraction of normal kidney function})$$

where f_e is the fraction of active drug excreted unchanged in urine.

CYP3A4 and P-Glycoprotein in Drug Absorption and Metabolism

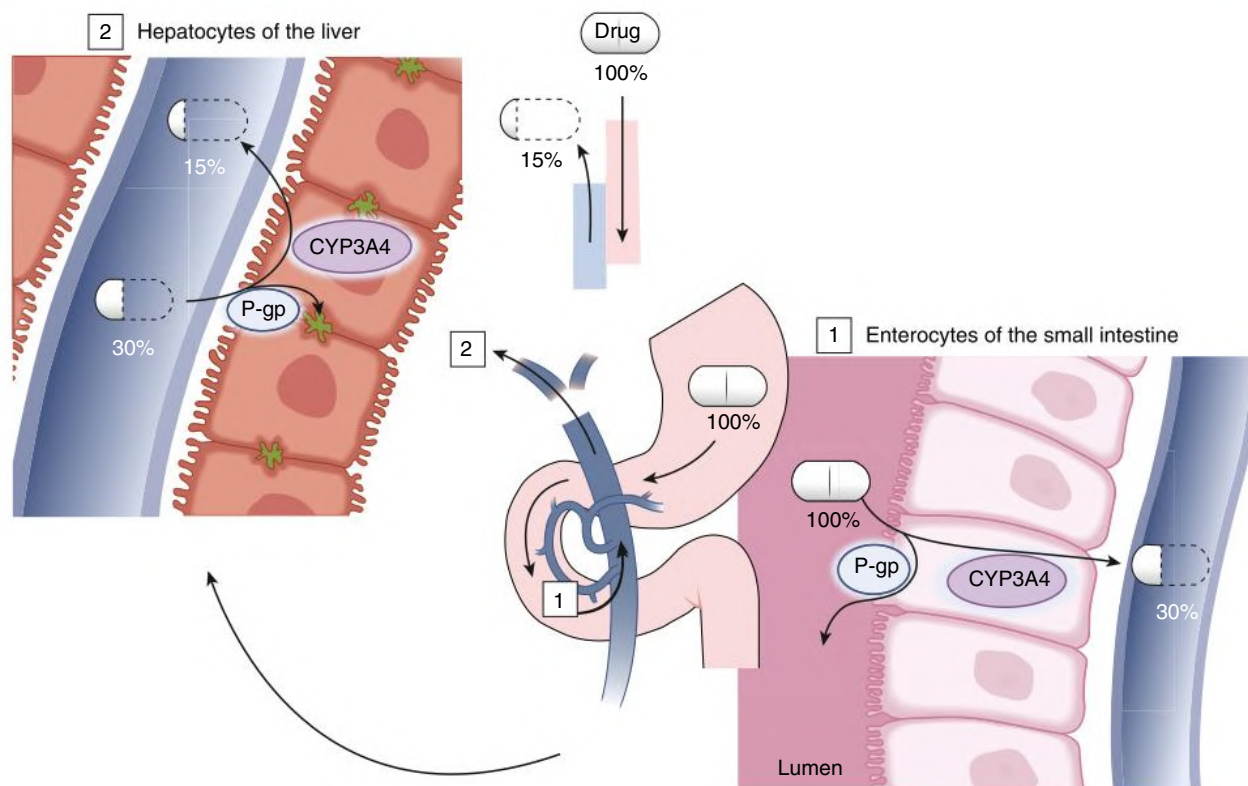


Fig. 77.4 CYP3A4 and P-Glycoprotein (*P-gp*) in Drug Absorption and Metabolism. Schema for absorption and metabolism of drugs that are substrates for CYP3A4 and *P-gp*. Overall, approximately 15% of ingested drug reaches the systemic circulation. (1) Enterocyte: *P-gp* on the apical surface of enterocytes prevents drug absorption, maintaining drug in the gastrointestinal lumen; CYP3A4 in enterocytes metabolizes drug. The net effect is that approximately 30% of ingested drug reaches the portal circulation. (2) Hepatocyte: *P-gp* on the cell surface prevents drug entry into hepatocytes; CYP3A4 in hepatocytes metabolizes the drug. The net effect is that drug entry to the systemic circulation is further reduced.

Once a daily dose reduction factor has been determined, the clinician must decide on a dose reduction method. Two methods are used, either alone or in combination (Fig. 77.5).

Interval Method

Using this method, the usual dose is administered at prolonged intervals where the extent of prolongation is similar to the extent to which GFR is reduced. This method is particularly useful when peak blood concentrations are important for efficacy (e.g., aminoglycosides) and when therapeutic concentrations need to be achieved early (oseltamivir). If this method is used, practical dose intervals should be recommended rather than inconvenient or complex intervals.¹⁵

Dose Method

Using this method, a smaller dose is given at the usual interval. This method is common, especially when peak concentrations are less critical for efficacy. If this method is used, clinicians should consider the availability of smaller dose formulations and the ability of the patient to accurately and safely divide available dosage forms.

Combination Method

Sometimes, especially for drugs with a narrow therapeutic index, for which tight control of concentrations is required (e.g., digoxin), a combination of the dose and interval methods is used.

Ongoing Assessment

Even with appropriate dose modification in CKD patients, clinicians should always remain vigilant and closely monitor the response to therapy to guide dose titration.

Therapeutic Drug Monitoring

Therapeutic drug monitoring can provide objective information to guide dosing and is valuable for drugs with a narrow therapeutic window such as aminoglycosides, glycopeptides, digoxin, lithium, antiepileptics, and immunosuppressants (see Table 77.6). Assays usually measure total blood concentrations and may significantly underestimate plasma concentrations of the active or free concentration of the drug.

Clinical Response

Ultimately, clinical response should influence the need to modify doses. Doses should be carefully titrated according to response and adverse effects. Measures of response include blood pressure for antihypertensives, blood glucose concentration and hemoglobin A_{1c} (HbA_{1c}) for oral hypoglycemics/insulin, and kidney function for toxicity from ACE inhibitors, and ARBs and urine output for loop diuretics.

Methods of Dose Reduction

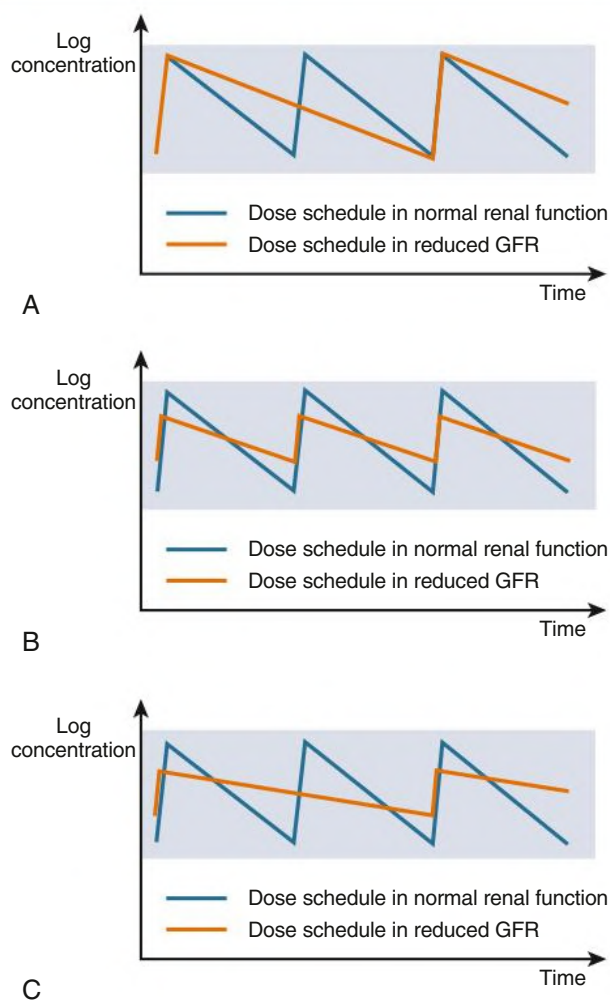


Fig. 77.5 Methods of Dose Reduction. (A) Interval method. (B) Dose method. (C) Combination method. *GFR*, Glomerular filtration rate.

EXTRACORPOREAL DRUG LOSSES

Failure to consider dialysis drug clearance may significantly reduce drug efficacy.^{5,7} Alternatively, dialysis may be used in overdose to assist drug removal (see [Chapter 103](#)). For example, some β -blockers have high dialysability and so are efficiently removed by dialysis (e.g., atenolol, acebutolol, metoprolol).¹⁶ In contrast, β -blockers with low dialysability are bisoprolol and propranolol. Studies of drug clearance with KRT modalities have often used variations in dialysis technique that make it difficult to compare results. Many studies before the 1990s report data from standard hemodialysis (HD) with low-flux membranes, which are less efficient at drug removal than the high-flux membranes now widely used. Many recommendations report the need for supplemental doses, although clinically, supplemental doses are rarely used, especially if less than 30% of the drug is cleared or the drug has a wide therapeutic window. Rather, drugs known to be significantly cleared by dialysis should be administered after dialysis ([Table 77.7](#)). When drugs are given in multiple daily doses, at least one of the daily doses should be administered soon after the completion of dialysis.

TABLE 77.6 Therapeutic Drug Monitoring in CKD

Drug	Therapeutic Range and When to Draw Sample
Aminoglycosides (24-h Dosing)	
	Peak (30 min after a 30-min infusion) to ensure adequate peak concentration
	Trough to guide time of next dose (2–3 half-lives post-dose), noting that half-life will be increased in approximate proportion to degree of reduction in GFR
Gentamicin	Peak: >10 mg/L Trough: Depends on time after dose 0.5–2 mg/L
Tobramycin	Peak: >10 mg/L Trough: Depends on time after dose 0.5–2 mg/L
Amikacin	Peak: >30 mg/L Trough: depends on time after dose 1.5–6 mg/L
Immunosuppressants	
Cyclosporine	C_0 (trough): 150–250 $\mu\text{g/mL}$ C_2 (2 h after dose): 1200–1500 $\mu\text{g/mL}$ $\text{AUC}_{0-4} >4400 \mu\text{g/mL/h}$
Tacrolimus	C_0 (trough): 4–12 $\mu\text{g/mL}$
Sirolimus	C_0 (trough): 5–15 $\mu\text{g/mL}$
Antiarrhythmics	
Digoxin	0.8–2.0 $\mu\text{g/mL}$ (trough at least 6 h after dose)
Lidocaine	1–5 $\mu\text{g/mL}$ 8 h after IV infusion starts or is changed
Antipsychotics	
Lithium	Acute: 0.8–1.2 mmol/L (trough) Chronic: 0.6–0.8 mmol/L (trough)
Antiepileptics	
Carbamazepine	4–12 $\mu\text{g/mL}$ (trough before administration)
Phenytoin	10–20 $\mu\text{g/mL}$ (trough before administration)
Free phenytoin	1–2 $\mu\text{g/mL}$ (trough before administration)
Phenobarbital	15–40 $\mu\text{g/mL}$ (trough before administration)
Valproic acid	40–100 $\mu\text{g/mL}$ (trough before administration)
Vancomycin	Trough: 15–20 mg/L (12 or 24 h trough)

Target levels are dependent on assay methodology and clinical context. *CKD*, Chronic kidney disease; *GFR*, Glomerular filtration rate; *IV*, Intravenous.

Hemodialysis

Hemodialysis drug clearance occurs mainly by passive diffusion from blood (high concentration) to dialysate fluid (low concentration) but can also occur by convectional movement of dissolved drug in ultrafiltrated plasma.^{7,17} The efficiency of drug removal depends on its physiochemical properties. As molecular size decreases (typically <500 kDa) and water solubility increases, drug removal increases. Conversely, as protein binding and V_D increase, the percentage of drug contained in plasma falls and dialysis clearance decreases. HD factors influencing drug removal include membrane type and surface area, blood and dialysate flow rates, and dialysis frequency and duration. HD sometimes removes drug from plasma faster than it can redistribute from tissue, so drug concentrations determined from samples drawn soon after beginning or completion of HD may be artificially low. Samples should preferably be drawn before dialysis or about 1 to 2 hours after to allow drug redistribution from tissues into plasma.

TABLE 77.7 Dialysis Clearance of Drugs

Drugs Significantly Cleared by Hemodialysis That Require Administration After Dialysis or a Supplement After Dialysis	Drugs Not Significantly Cleared by Hemodialysis and That Do Not Require Administration After Dialysis or a Supplement After Dialysis or That Must Be Administered at Specific Times Independent of Hemodialysis
Analgesics Morphine 6-glucuronide (in toxicity) Aspirin, high dose (contraindicated)	Anemia Erythropoietins Iron
Antibiotics Aminoglycosides Amikacin Gentamicin Tobramycin Cephalosporins Cefotaxime Cefazolin Ceftazidime Carbapenems Imipenem Meropenem Metronidazole Penicillins Amoxicillin Ticarillin Piperacillin Fluoroquinolones Ciprofloxacin Glycopeptides Vancomycin (high-flux dialyzers) Teicoplanin Miscellaneous antibiotics Ethambutol Cotrimoxazole	Analgesics Paracetamol Fentanyl Oxycodone NSAIDs Antibiotics Penicillins Flucloxacillin Fluoroquinolones Moxifloxacin Rifamycins Rifampin Rifabutin Glycopeptides Vancomycin (oral administration) Tetracyclines Tetracycline Doxycycline Minocycline
Antifungal Fluconazole	Antifungals Amphotericin Voriconazole Ketoconazole Itraconazole
Antivirals Acyclovir Cidofovir Famciclovir Foscarnet Ganciclovir Ribavirin Valganciclovir Zidovudine	Antiviral Amphotericin Antiepileptic Carbamazepine
Antineoplastics Cyclophosphamide Methotrexate	Cardiovascular Agents Amiodarone Perhexiline Nitrates
Anticoagulant Lepirudin Dabigatran (in overdose)	Anticoagulants Heparin Warfarin LMWH Rivaroxaban Apixaban
Antiepileptics Gabapentin Pregabalin Levetiracetam	Antiplatelet Drugs Low-dose aspirin Clopidogrel Dipyridamole Immunosuppressants Azathioprine Cyclosporine and tacrolimus Mycophenolate Prednisolone

TABLE 77.7 Dialysis Clearance of Drugs—cont'd

Drugs Significantly Cleared by Hemodialysis That Require Administration After Dialysis or a Supplement After Dialysis	Drugs Not Significantly Cleared by Hemodialysis and That Do Not Require Administration After Dialysis or a Supplement After Dialysis or That Must Be Administered at Specific Times Independent of Hemodialysis
Cardiovascular Agent Sotalol	Sirolimus and everolimus T cell–depleting antibodies
Antidiabetic Metformin (in overdose)	Rituximab Anti-CD25 antibodies
Vitamins Water-soluble B, C, and folate	Antidiabetics Sulfonylureas
Psychotropic Lithium	Musculoskeletal Agents Bisphosphonates
	Antiepileptics Sodium valproate Carbamazepine
	Antidepressants SSRIs (administer in morning)
	Vitamins Fat-soluble vitamins A, D, E, K

LMWHs, Low molecular weight heparins; NSAIDs, nonsteroidal antiinflammatory drugs; SSRIs, selective serotonin reuptake inhibitors.

Peritoneal Dialysis

Many drug properties that affect removal by HD also apply to peritoneal dialysis (PD), but PD is usually less efficient.¹⁸ For significant removal by PD, the drug must have a very low V_D and low protein binding. For most drugs, there is little evidence of significant removal during chronic PD. A few studies have examined drug clearance from automated PD that uses large volumes of short dwells overnight, often accompanied by one or more longer daytime dwells. Clearance of some drugs via automated PD is increased because of the increased drug concentration gradient between blood and dialysate. Increased drug dialyzability may occur with increased peritoneal dialysate flow rates or during peritonitis, although overall efficiency of drug and metabolite clearance is often reduced by peritonitis because of scarring of the peritoneal membranes, thereby reducing effective surface area.

Continuous Kidney Replacement Therapy

Drug clearance by continuous KRT (CKRT) with hemofiltration, HD, or hemodiafiltration differs from that by intermittent HD.¹⁹ Relying on continuous ultrafiltration of plasma, CKRT can remove large quantities of ultrafilterable drug. CKRT generally uses membranes with larger pore sizes and involves convective transport of solute. Hence, CKRT allows the passage of larger molecules (up to 5000 Da). A large V_D and protein binding still prevents removal by CKRT. Protein binding and the filtration rate determine the rate of removal. A series of sieving coefficients is available that allows calculation of the amount of drug actually lost if the ultrafiltration flow rate is known.⁵ The sieving coefficient is the ratio of drug concentration in the ultrafiltrate to the prefilter plasma drug concentration. The closer the sieving coefficient is to 1.0, the more it passes across the filter. There are few detailed studies of drug clearance with use of these methods, and clinicians must rely on estimates from HD, known physiochemical properties, and clinical response.

POLYPHARMACY AND DEPRESCRIBING

Patients receiving dialysis typically receive 10 to 12 medications. Polypharmacy increases the risk of medicine-related problems, including adverse drug reactions and nonadherence. Polypharmacy may be systematically addressed through “deprescribing,” an evidence-based,²⁰ patient-centered²¹ process that enables identification and elimination of unnecessary or inappropriate medications. In elderly patients on dialysis, consideration for deprescribing should be given to quinine, diuretics, α_1 -blockers, proton pump inhibitors, and statins.²² In older adults with CKD, medication optimization and deprescribing should focus on proton pump inhibitors, oral hypoglycemic agents, and statins.²³

Electronic Medical Record

The implementation of electronic medical record (EMR) should improve health outcomes of kidney patients through improved clinical decision making, medication safety, patient education, and quality improvement and research. EMR should incorporate the design and development of clinical care pathways for patients with kidney disease. Medication therapy assessment and monitoring tools should be a part of computerized provider order entry (CPOE) for medication safety to provide evidence-based advice about drug interactions and dose adjustments for patients with kidney disease or transplantation. Furthermore, interactive educational programs and tools to aid medication adherence should be integrated into the patient EMR to improve health literacy and patient outcomes in vulnerable populations. Moreover, EMR data could be used in pharmacoepidemiology research to provide the best available evidence for treatment outcomes in the patient population that is frequently excluded from the randomized controlled studies.²⁴

SELF-ASSESSMENT QUESTIONS

1. What percentage of the dose used in patients with normal kidney function should be given for a hypothetical drug that is 80% renally cleared unchanged in urine to a person with 25% of normal kidney function?
 - A. 20%
 - B. 30%
 - C. 40%
 - D. 60%
 - E. 80%
2. Which of the following factors or drug characteristics does *not* increase the likelihood of drug removal by hemodialysis?
 - A. High-flux membranes
 - B. Low protein binding
 - C. Low volume of distribution
 - D. Large molecular weight
3. Which of the following drugs does *not* require clinically relevant dose modification in patients with reduced GFR?
 - A. Amoxycillin
 - B. Ribavirin
 - C. Digoxin
 - D. Sotalol
 - E. Enoxaparin
4. Which of the following drug combinations is *least* likely to cause clinically significant drug interactions?
 - A. Adding clarithromycin to the regimen of a patient on sirolimus
 - B. Adding phenytoin to the regimen of a patient on prednisolone
 - C. Adding fluconazole to the regimen of a patient on tacrolimus
 - D. Adding roxithromycin to the regimen of a patient on cyclosporin
 - E. Adding verapamil to the regimen of a patient on everolimus

REFERENCES

1. Manley HJ, McClaran ML, Overbay DK, et al. Factors associated with medication related problems in ambulatory hemodialysis patients. *Am J Kidney Dis*. 2003;41:386–393.
2. Cervelli MJ, ed. *The Renal Drug Reference Guide*. MJC Pharma; 2007.
3. Ashley C, Dunleavy A, eds. *The Renal Drug Handbook*. 5th ed. CRC Press; 2018.
4. Aronoff GR, Berns JS, Brier ME, eds. *Drug Prescribing in Renal Failure (Dosing Guidelines for Adults and Children)*. 5th ed. Philadelphia: American College of Physicians; 2007.
5. Johnson CA. *Dialysis of Drugs*. Verona, WI: CKD Insights; 2009.
6. Meibohm B, Evans WE. Clinical pharmacokinetics and pharmacodynamics. In: Helms R, Quani D, Herfindal ET, eds. *Textbook of Therapeutics: Drug and Disease Management*. Philadelphia: Lippincott Williams & Wilkins; 2006:1–20.
7. Matzke GR, Comstock TJ, et al. Influence of renal function and dialysis on drug disposition. In: Burton ME, Shaw LM, Schentag JJ, eds. *Principles of Therapeutic Drug Monitoring*. 4th ed. Philadelphia: Lippincott Williams & Wilkins; 2006.
8. Zhang Y, Benet LZ. The gut as a barrier to drug absorption: combined role of cytochrome P450 3A and P-glycoprotein. *Clin Pharmacokinet*. 2001;40:159–168.
9. Frost RW, Lasseter KC, Noe AJ, et al. Effects of aluminum hydroxide and calcium carbonate antacids on the bioavailability of ciprofloxacin. *Antimicrob Agents Chemother*. 1992;36:830–832.
10. Reidenberg MM, Drayer DE. Alteration of drug-protein binding in renal disease. *Clin Pharmacokinet*. 1984;9(suppl 1):18–26.
11. Dreisbach AW, Lertora JLL. The effect of chronic renal failure on hepatic drug metabolism and drug disposition. *Semin Dial*. 2003;16:45–50.
12. Lam YW, Banerji S, Hatfield C, Talbert RL. Principles of drug administration in renal insufficiency. *Clin Pharmacokinet*. 1997;32:30–57.
13. Reidenberg M. Kidney function and drug action. *N Engl J Med*. 1985;313:816–817.
14. Elbarbry FA, Marfleet T, Shoker AS. Drug-drug interactions with immunosuppressive agents: review of the *in vitro* functional assays and role of cytochrome P450 enzymes. *Transplantation*. 2008;85:1222–1229.
15. Jones TE. Oseltamivir: current dosing recommendations reduce the therapeutic benefit in patients with mild to moderate renal function and/or large body mass: a review of the literature with recommendations to optimize dosing, including the use of therapeutic drug monitoring. *Ther Drug Monit*. 2021;43(1):103–107.
16. Weir MA, Dixon SN, Fleet JL, et al. beta-Blocker dialyzability and mortality in older patients receiving hemodialysis. *J Am Soc Nephrol*. 2015;26(4):987–996.
17. Lee CS, Marbury TC. Drug therapy in patients undergoing haemodialysis (clinical pharmacokinetic considerations). *Clin Pharmacokinet*. 1984;9:42–66.
18. Paton TW, Cornish WR, Manuel MA, Hardy BG. Drug therapy in patients undergoing peritoneal dialysis: clinical pharmacokinetic considerations. *Clin Pharmacokinet*. 1985;10:404–426.
19. Bohler J, Donauer J, Keller F. Pharmacokinetic principles during continuous renal replacement therapy: drugs and dosage. *Kidney Int*. 1999;56(suppl 72):S24–S28.
20. Scott IA, Hilmer SN, Reeve E, et al. Reducing inappropriate polypharmacy: the process of deprescribing. *JAMA Intern Med*. 2015;175(5):827–834.
21. Todd A, Jansen J, Colvin J, McLachlan AJ. The deprescribing rainbow: a conceptual framework highlighting the importance of patient context when stopping medication in older people. *BMC Geriatr*. 2018;18(1):295.
22. McIntyre C, McQuillan R, Bell C, Battistella M. Targeted deprescribing in an outpatient hemodialysis unit: a quality improvement study to decrease polypharmacy. *Am J Kidney Dis*. 2017;70(5):611–618.
23. Triantafylidis LK, Hawley CE, Perry LP, Paik JM. The role of deprescribing in older adults with chronic kidney disease. *Drugs Aging*. 2018;35(11):973–984.
24. Strippoli GF, Craig JC, Schena FP. The number, quality, and coverage of randomized controlled trials in nephrology. *J Am Soc Nephrol*. 2004;15:411–419.